

Bus Trip (trip)

In Italy there are N cities numbered from 0 to $N - 1$. Luca lives in city S , but for the holidays he decided to go visit city E . Being just a student he doesn't want to spend a fortune in plane tickets, so he decided to have a nice **bus trip**!



Luca used a Bash script to download the complete schedules of all the bus companies in the country. He collected M bus routes, and for each of those he has collected the following four details:

1. the starting city;
2. the time of bus departure;
3. the arrival city;
4. the time of bus arrival.

Assuming that Luca is located in the city S at time $t = 0$, help him compute the earliest possible time he can reach the city E to enjoy his holiday, by using as many buses as needed.

 Among the attachments of this task you may find a template file `trip.*` with a sample incomplete implementation.

Input

The first line contains the integer N . The second line contains two integers S and E . The third line contains the integer M . Each of the following M lines describe a bus route, and is formed by 4 integers, respectively: the starting city s , the starting time t_0 , the arrival city e , the arrival time t_1 .

Output

You need to write a single line with an integer: the earliest time when Luca can reach E starting from S . If it's not possible to reach city E with the given schedules, write "IMPOSSIBLE" without quotes.

Constraints

- $1 \leq N \leq 50\,000$.
- $1 \leq M \leq 100\,000$.
- There is at most 1 bus route between two cities.
- For each bus route (s, t_0, e, t_1) , it will hold: $0 \leq s, e < N$, $s \neq e$, and $0 \leq t_0 \leq t_1 \leq 10^9$.
- Luca can catch a bus that leaves at time t_0 if he arrives in its starting city at any time $t \leq t_0$.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- Subtask 1 (0 points)

Examples.
- Subtask 2 (20 points)

$N \leq 10, M \leq 30$.
- Subtask 3 (30 points)

There is one and only one possible path that connects any two cities.
- Subtask 4 (30 points)

$N \leq 1000, M \leq 3000$.
- Subtask 5 (20 points)

No additional limitations.

Examples

input	output
3 0 2 4 2 0 1 5 0 1 1 3 1 3 2 5 0 0 2 6	5
3 0 2 3 0 0 1 5 0 1 1 4 1 3 2 5	IMPOSSIBLE

Explanation

In the **first sample case**, Luca can catch the bus that leaves city 0 at $t = 1$, arrive at city 1 at $t = 3$, and then leave again at $t = 3$ towards city 2, arriving at $t = 5$.

In the **second sample case** Luca has no way to reach city E with buses only.